Data Science Lab

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Import Required Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler
from keras.models import Sequential
from keras.layers import LSTM, Dropout, Dense
```

Load Training Data

```
from google.colab import files
uploaded = files.upload()

dataset_train = pd.read_csv('Google_Stock_Price_Train.csv')
training_set = dataset_train.iloc[:, 1:2].values
print(f"Training data shape: {training_set.shape}")
```

Feature Scaling

```
sc = MinMaxScaler(feature_range=(0, 1))
training_set_scaled = sc.fit_transform(training_set)
```

Create a Data Structure with 60 Timesteps and 1 Output

```
X_train, y_train = [], []
for i in range(60, len(training_set_scaled)):
    X_train.append(training_set_scaled[i-60:i, 0])
    y_train.append(training_set_scaled[i, 0])

X_train, y_train = np.array(X_train), np.array(y_train)
X_train = np.reshape(X_train, (X_train.shape[0], X_train.shape[1], 1))
print(f"X_train shape: {X_train.shape}")
print(f"y_train shape: {y_train.shape}")
```

```
X_train shape: (1198, 60, 1)
y_train shape: (1198,)
```

Build the RNN with LSTM Layers

```
regressor = Sequential()
regressor.add(LSTM(units=50, return_sequences=True, input_shape=(X_train.shape[1], 1)))
regressor.add(Dropout(0.2))
regressor.add(LSTM(units=50, return_sequences=True))
regressor.add(Dropout(0.2))
regressor.add(LSTM(units=50, return_sequences=True))
regressor.add(Dropout(0.2))
regressor.add(LSTM(units=50))
regressor.add(Dropout(0.2))
regressor.add(Dense(units=1))
regressor.compile(optimizer='adam', loss='mean_squared_error')
```

Train RNN

```
regressor.fit(X_train, y_train, epochs=100, batch_size=32)
```

```
Epoch 1/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 12s 41ms/step - loss: 0.1266
Epoch 2/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 18ms/step - loss: 0.0071
Epoch 3/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 1s 17ms/step - loss: 0.0055

Epoch 98/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 1s 27ms/step - loss: 0.0017
Epoch 99/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 1s 16ms/step - loss: 0.0016
Epoch 100/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 1s 17ms/step - loss: 0.0017
<keras.src.callbacks.history.History at 0x7ca90c72b050>
```

Load the Test Dataset

```
uploaded = files.upload()

dataset_test = pd.read_csv('Google_Stock_Price_Test.csv')
real_stock_price = dataset_test.iloc[:, 1:2].values

dataset_total = pd.concat((dataset_train['Open'], dataset_test['Open']), axis=0)
inputs = dataset_total[len(dataset_total) - len(dataset_test) - 60:].values
inputs = inputs.reshape(-1, 1)
inputs = sc.transform(inputs)
```

```
X_test = []
for i in range(60, 60 + len(dataset_test)):
    X_test.append(inputs[i-60:i, 0])
X_test = np.array(X_test)
X_test = np.reshape(X_test, (X_test.shape[0], X_test.shape[1], 1))
```

Predict and Visualize Results

```
predicted_stock_price = regressor.predict(X_test)
predicted_stock_price = sc.inverse_transform(predicted_stock_price)

plt.figure(figsize=(12, 6))
plt.plot(real_stock_price, color='red', label='Real Google Stock Price')
plt.plot(predicted_stock_price, color='blue', label='Predicted Google Stock Price')
plt.title('Google Stock Price Prediction')
plt.xlabel('Time')
plt.ylabel('Stock Price')
plt.legend()
plt.show()
```

